

HR Analytics
Prof. Santosh Rangnekar, Professor
Department of Management Studies
Indian Institute of Technology, Roorkee
Dr. Abhishek Singh, Assistant Professor, OB & HR
Indian Institute of Management, Rohtak
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Lecture 02: HR Data Preparation & Development of HR metrics

Dear participants, in first session of HR analytics, we had learned the basics about the HR analytics. So, what this analytics is, what type of questions that can be answered, in what situations a manager can take a decision, right. So, these are the things that we had learned in the first session. In this session, we will learn how to prepare this HR data, right, data which you need to collect and then you have to transform, you have to make some changes before analyzing the data. So, here we will learn how this data should be collected, what type of preparation we should do before collecting the data and after collecting the data what we should do. So, right here so that data preparation thing is what we will understand right in this session.

So, let us start. So, this is the content of your data. So, questions related to the data that you should ask. So, what are the questions that you should ask related to your data? So that you will see your data is accurate, reliable and applicable.

So, what type of questions that you should ask in first thing that is what we will discuss. Next thing that we will discuss about the data measurement, how this data should be measured, whatever data that you are And third thing that data preparation, how this data should be prepared, what are the things that we should consider. The moment we have decided on the measurement of data, then, we should think about how we should develop the HR matrix through which we will collect and analyze the data, right? And what are the criteria for good HR activity matrices, right? Any activity that is taking place, so what are the criteria? So, these are the things that we will discuss in today's session. So, let us start with the questions that a manager should ask to himself or the team which is going to deal with the data, right? So, this is the thing that I always say. No matter how much you trust your quant or your data, do not stop asking them tough questions.

It does not matter how much you rely on your data or your numbers, but still, you ask the tough questions, right? So, what are those tough questions that you should ask related to your data, that is what we will discuss. But this is the thumb rule for the data preparation. It does not matter how much you trust your number. If you have a 100 percent trust also,

then also you should ask certain questions. So, what are those questions, that is what we will discuss, right.

The first question that you should ask related to the data is what was the source of your data, from where your data was collected, right? So, as you know, there could be many sources of the data. Some data that you will get is related to the HR internal right, which is there within the organization, and for some matrices, you have to collect the data from outside also. So, outside of the case of salary right, in the case of salary if you have to compare whether whatever salary that you are giving is equivalent to your competitors or not. So, in that case you are collecting the salary or CTC data from outside the organization. So, you need to understand from which source through which source you have collected the data if that source is not reliable then reliability of the data may be questionable right so that is why i hope you can understand now why the source of data is important right.

In a similar way, in internal also, sometimes some data is not available within the HR department. For example, the performance rating of all employees is available to you, but the sales data, how much sales each individual has achieved, that data may not be available to you, right? So, in that case, if you have to correlate performance rating and sales data, right, you have to correlate these things, then also you have to take that sales data from the marketing or sales department. So whether, although it is internal data but still you have to ask what is the source of that data, right. So in, so data can be collected through internal sources or external sources, right. So that is why it is important to ask the first question that is the what is the source of data.

So if you believe your source is reliable then you can rely on the data also. If you believe that source is not reliable, right, from where you have collected the data, then your data is questioned, may be questionable, and your results through that data may be questionable. So, that is what you can put it as a limitation if you do not have any other source to collect the reliable data. If that is the only source and you are not sure about it, you can collect data, but you can put it as a limitation of your analysis, right. The second question that you can ask is how well the sample, whatever sample you have collected, because it is very difficult to do the analysis on the entire population.

So I have already given the example of this external data of salary. So if you have to compare the salary for one job position with your competitor then it is not possible to collect the salary structure or CTC information about all employees who are working with your competitor right for this kind of analysis you have to take some sample from if five competitors are there so from each competitor you can take five employees six employees, it is up to you, right. So you have to think about whether this sample is able

to explain the population characteristics or not. If sample characteristics and population characteristics are not matching then you cannot make the prediction about the population characteristics, right. I hope you understand what I am saying.

Sample characteristics and population characteristics should be same. If both are not the same, then you will not be able to make predictions about the population characteristics based on the sample analysis, right? So that similarity that you have to check, right. So before the data analysis if you whatever data you are collecting then you should ensure that whatever sample you have taken for the analysis that same samples should have the similarity to the population characteristics so that you can predict something about the population based on the sample results that the sample that you have collected right. The third question that you should ask is whether your data distribution includes outliers. So, if outliers are there then so then if you answer yes or no, outlier I hope you understand, like whatever is the normal data.

So, CTC that you are collecting for the manager position. In all organization that you said 50k per month is there. But in one of the organizations, it is 2 lakh for one manager, and you have collected the data of 6 managers. So, this 2 lakh will be the outlier. So, values which are very low or extremely high as compared to other samples, right?.

So that is what you can say outlier. So if question is yes, your data include the outlier, the next question that you should ask, how did they affect your result? Because if you will include the outlier, you will not understand how it impacts the result. Analysis may be misleading, right? So, if you do not want to give misleading result or if you do not want to make inference, misleading inference based on your data analysis then think about the outlier, right. What assumptions are behind your data analysis? So, when before data analysis you need to clarify what are the assumptions that your data analysis is having. That is what clearly you have to define that particular thing, what are the assumptions.

Why did you decide that particular analytical approach? So whenever you will analyze the data, so before analyzing you have to take one approach. So you can select one of the approaches from descriptive statistics, diagnostic statistics, predictive statistics or prescriptive statistics. So whichever analytical approach that you have selected from this analytical approach is descriptive, diagnostic, prescriptive, and predictive. So, you have to give a reason behind it, what is the particular reason behind this particular approach that you have selected, and what are the alternatives available. If you are not selecting this particular approach, then what alternatives are available? So, these are the questions that you should keep on asking.

So, here you can see some of the questions are about the sources of the data; some questions are about the data analysis, and some questions are about the representation, right, and approach. So, these are the questions that you must ask yourself before analysing the data in order to increase the validity and reliability of your data. Next aspect, whatever data that you are going to collect, HR related data. Next, how you will measure that data. How you will measure? You will take it as a number, absolute value like number of applicants, number of applicants, right.

So, have you decided to take it as a number, right or in the form of ratio or in the form of percentile, right. So, that you have to decide. So, ratio that also you can decide how many applicants are there, number, ratio, male, to female ratio. So, what is the male to female ratio is there? So, this data you can present it in the form of percentage like 24 percentage males are there and 64 percentage female are there and 2 is to 3 ratio also you can present it. So, it is up to you which type of measurement that you want to do for a particular data.

In the same way, percentile, would you like to present the data in the percentile format? So, here is the result that you are declaring, right? that is what result you can declare, the way you would have seen the CAT result, the common admission test result is there, that is also presented in the form of the percentile. So, if you are getting the 99-point percentile, it means 0.1 per cent of students are ahead of you. So, that tells you what your position is among all students. So, if you are having 98 percentile, it means 2 percent students are before you, right.

So, that is how you can present this data. So, result data that you can present in the form of the percentile. The fourth form of data presentation is what you will see that is the correlation information, right? Some data if you will present without in just in number format may not give you the very big meaning, but when you will present in the correlation format. So, one example that I have already given here is a second example that I can give a performance rating along with the sales target achieved. So, if performance ratings are high and the sales target achieved is also high, that data gives you the meaning, right? So, here you will be able to see some patterns.

Same thing when you will see leaders behaviour and employee engagement. So, leader behaviour can be presented in the form of whether the leader respects his or her subordinate or not. If the leader is respecting then you can see the engagement level. So, on a 1 to 7 level scale, you can measure the engagement and on a 1 to 7 scale, you can measure the leader's behavior. So, if leaders behavior is employee oriented, right, relationship oriented and then you can see the engagement.

So, in the case of relationship oriented, respect oriented, task oriented. So, in which

orientation leaders' behaviour has an impact on the engagements. When we are presenting this data in the form of correlation, that will give you a lot of information. So, second aspect that you need to decide about the data, how you will measure this data? Whatever HR data that you are going to collect, how you will measure? Will you measure in the form of absolute number? Will you measure in the form of ratio or percentage or will you present the data in the form of percentile and will you present the data in the correlated information, right. So, because each data has a nature and how it should be presented, it will give you more meaning and more information.

And it will increase your ability to analyze more. So that is what you need to think, and then you have to decide which type of measurement you will do with the particular data, right? So, I hope you understand the importance of this data measurement, right? So now data preparation, so what are the aspects of the data that is what you have to check. So, regarding the reliability of data I already said, from which source you are collecting the data, if that source is not reliable, then your data reliability also may be questionable. Same thing with the data accuracy, right, how accurate that data is. So, that is the thing that you can see.

Data completeness. whatever data that you have collected for the analysis whether it is complete or not, data variation how much variation is there in the data, data applicability is there and missing data information that is what you can check. So, these are the parameter you need to check the data that you have collected. So how reliable it is, how accurate it is, how complete it is, how much applicable it is, right. So these are the parameter on which you should check the data preparation, right. So if any one of the parameter that you see there is a problem then you should raise the red flag for that particular issue.

So that before the analysis itself you can correct that particular data and you can transform that data you can add something you can check the sources you can do something but after the analysis report is prepared decision is taken it may be difficult so before data collection, and after the data collection these issues should be raised I always suggest before the data collection, check the reliability and validity of the questionnaire that you have developed and the reliability and validity of the source from which source that you are going to collect the data. So, if source is reliable go ahead and take the data. If you have a question about the source think again and again and think about some other alternatives also. If alternatives are there then you can go ahead with the alternatives. So, once you have thought these all things, then you have to think how to develop the HR matrices, right, how to develop the HR matrices.

So, one thing that you can remember, every KPI is a matrix, every KPI. So, you would

have seen KPI in the organization is having the number. There is no KPI which is not having the number. All KPIs are having the number, right. For example, engagement, employee engagement you need to increase that is your KRA, right.

For this KPI you have decided some of the activities. So, one activity for employee engagement that you have decided. So, one HR department has decided for employee engagement, one activity that you will give some information about job activities. Every day to the each employee. So, how many activities? One related to one activity.

So, that one activity information is given to the employee or not. So, that is there, so that one is the KPI. In the same way for a recruitment, so recruitment let us assume 20 employees that you need to hire in next one year for senior level. So, you have decided you will do multiple activities like you have to do the job posting, right, advertisement and then application screening that is what. So, for first month you have decided the job postings on a multiple sources.

So, your your KPI is one activity on internal source and one activity for external. So, every month you have to do two job posting, one on internal and one on external. So, KPI is having to your, KPI is one for internal and one for external. So, both are number. So, these numbers are the metric because they are measuring something, right.

So KPI that is what you can see something they are measuring. So, these all KPIs. So, whatever KPIs that during the goal setting process some KRAs would be there and in order to measure that KRA, progress towards that KRA some KPIs are there. So, that those numbers that you have decided. So, every number is a metrics whatever the number that you see in a KPI that is a metrics.

So you think about it. Now question comes how to develop the metrics. So I always say HR department if you are thinking about the HR department so these are the 6 activities that HR department does. What are those six activities? Recruitment and selection, learning and development, performance and compensation. So, these are the six activities.

You think about these activities. So, in upcoming sessions you will learn the process of each function. So, what is the recruitment process? What is the process of selection? What is the process of learning? What is the process of development? What is the process of compensation? What is the process of performance? So, these all processes you will learn in the upcoming sessions. So, the moment you have understood the processes, think about what are the challenges are there related to the each process. So, those challenges are there.

And then only you make a list of variables. So, in the case of recruitment, for example, in the case of recruitment, first challenge is employer branding, how strong your employer brand is, that you want to know, right. If you want to know how strong your employer brand is, right, so what you can do, how we measure the strength of employer brand, employer brand strength we measure through how many people have left your organization and how many people have joined your organization from the competitor, right. How many have joined to the competition and how many have left for the competition. So, right, so that is how you can see. So, for example, from your organization 10 people have joined your competitor.

And 12 people from the competitor have joined you so you can see two members additional member have joined you so it indicates that your employer brand is more stronger than the, your competitors. So, first what you did, first you identify the problem and then you thought which type of data will help you to understand the extent of that problem. So, you decided the leaving people and coming people that you will understand from the definition of that particular problem. So, you understand that particular concept then decide which data will be more applicable, collect the data and calculate it. So, related to each function that is what you have to do.

Related to each function you have to identify certain problems, challenges that manager is facing and then you make a list of variables, based on that collect the data, analyze it and make the decision. So, first you need to think about the function, under which function, which problem is there and then collect the data. Why I am focusing on problem? Because if you will collect the data without problem, then what is the use of that data, right? So, you have to make a decision in order to solve some problem in the organization. So, that is why I am saying you make an index matrix first related to those issues for which your organization is facing the problem. If attrition is not the problem in the organization, people are not leaving the organization, then why you want to waste your time to analyze the attrition? When attrition is zero in your organization, people are not leaving at all, then why you want to analyze? Some other problems are there, go ahead and analyze those problems rather than wasting your time in attrition because there has to be a problem.

If problem is there, then only you collect the data related to that, develop the matrices, analyze it and make a decision. Do not waste your time on those issues for which there is no problem in the organization, right. So, that is the first aspect, how you identified these are problems. Second step is divide these all problems as per the functions. So, all HR related problem that are there that you divide into 6 under these 6 functions related to various steps of these processes.

So, these steps and challenges that manager face and what are the steps of each process that is what you will learn in coming. So, what I am giving you whatever problem that manager is facing divide under these functions and then you can develop the matrices so that you will be able to categorize this is the recruitment matrices, this is the selection matrices, this is the training matrices, this is the development matrices, this is the performance matrices and this is the compensation matrices. So I hope you would have understood the categorization of data, categorization of these matrix also. So, I hope it would have been clear to you how to develop the matrices related to the HR. So, few criteria that is what we can learn, that we should keep in our mind while developing the matrices, so that we will be able to develop some good activity matrices.

So, first thing that we should remember, every matrices should, whatever we are developing it should be documented, right. It should not just you have developed and then you forgot, you calculated and you have forgot, no. Whatever metrics that you are developing, your document, and that process of documentation, I already said you divide all matrices into 6 categories, some metrics related to recruitment, compensation, selection, and performance, right? So, related to these all 6 functions you can divide and then you can document it. Second, matrices should have the narrative definition. So, whatever that you are going to capture, let us assume the percentage of absenteeism, absenteeism that is the you are calculating.

So, what is this total employee total employees absent number of employees who are absent divided by total employees and multiply by 100. So, on that particular day what is the absenteeism rate right. So, that is what that is what any you can calculate. So, you have to define these all things what do you mean by this absenteeism right how you are calculating it. So, this definition should be there explain the all unfamiliar things which people cannot understand.

Third parameter that I suggest for each metric there has to be a mathematical formula. Even if it is count also, right? Number that you are presenting that is also a formula. So, if it is even count also then also you have to have a mathematical formula for every metric. If metric is unable to measure something then that metric is not a metric, right? Metric should be able to calculate something.

So, for that you need to have a formula. If counting also, that is also giving you some number and based on that number you can interpret something. So, that is why counting also in mathematical formula. So, you should have the mathematical formula for every metric that you are developing related to these HR functions, right. Next criteria, every metric should provide the information as to what to calculate and what write what you have to calculate. That is what every metric should provide, and you should explain

clearly.

Every metric should include a description of why it is being captured. Whatever information that you are capturing through the metrics then you should explain why you are doing that. Because as I already said if you are collecting the data which is related to that there is no problem. Right, and if you do not have any plan to use that particular data in future in the planning process to solve or make the process better, then why collect that data? So, if you need that data then only you go ahead and collect the data. So, that is why this description is important related to each metrics why you are capturing that particular data and for what you will use it. Next criteria, you should categorize these all metrics, so that I have already explained how you can categorize in among all 6 categories.

Every metrics should indicate the official source of the raw data. So, your metric will calculate something, so the data from where that data will get it, so what is the source of the data that also should mentioned, so that anyone who is using that metric, that person should be able to understand what is the source of data and what is the reliability of that analysis. So, every metrics should explain the official source of the raw data. So, thank you. I hope you would have learned the basics of data preparation related to the HR. Thank you.